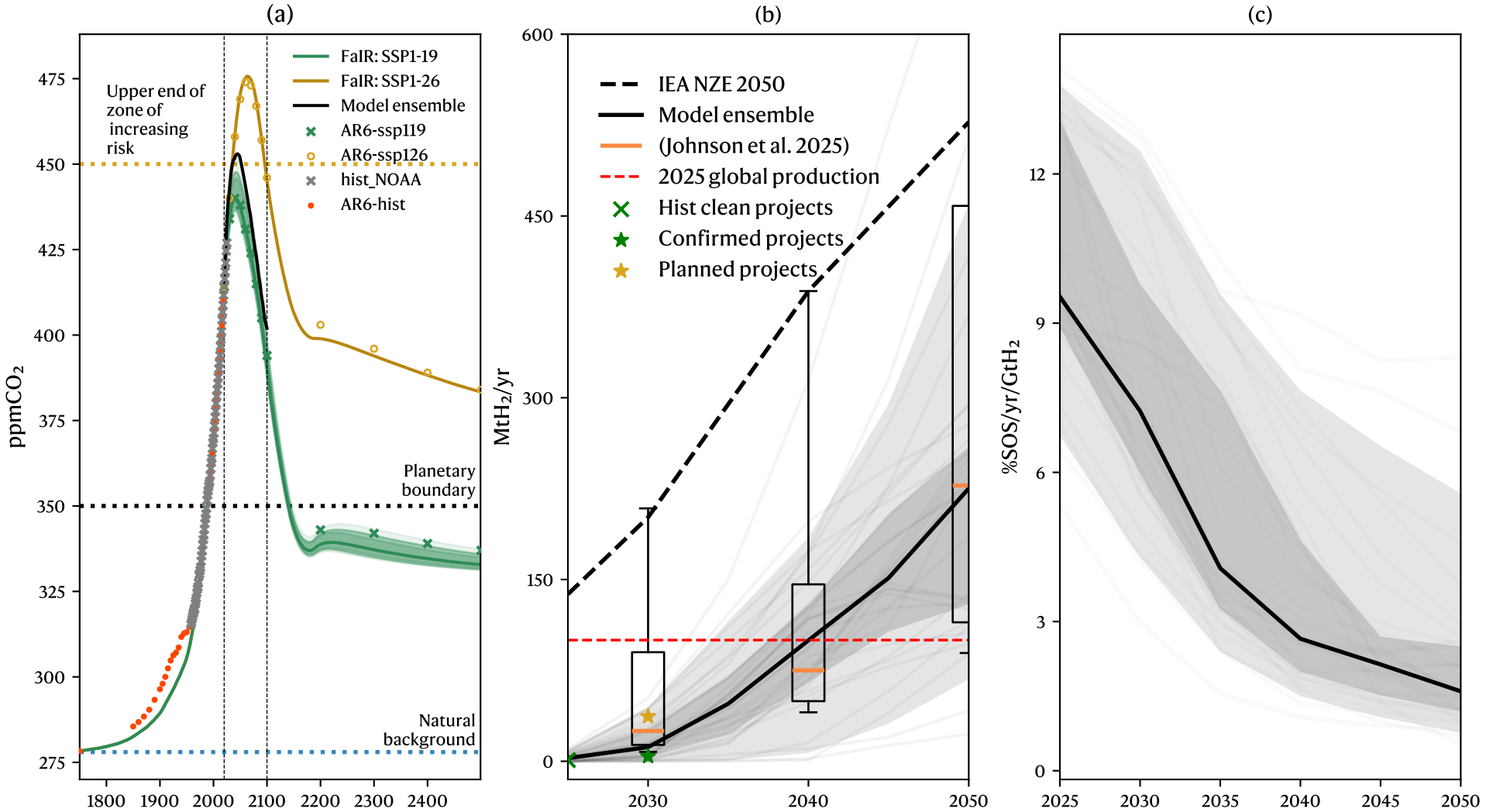
(a)
FaIR: SSP1-19
FaIR: SSP1-26
Model ensemble
AR6-ssp119
AR6-ssp126
hist_NOAA
AR6-hist
Upper end of zone of increasing risk
Planetary boundary
Natural background
ppmCO2
475
450
425
400
375
350
325
300
275
1800
1900
2000
2100
2200
2300
2400
(b)
IEA NZE 2050
Model ensemble
(Johnson et al. 2025)
2025 global production
Hist clean projects
Confirmed projects
Planned projects
MtH2/yr
600
450
300
150
0
2030
2040
2050
(c)
%SOS/yr/GtH2
12
9
6
3
0
2025
2030
2035
2040
2045
2050